

Think College **REPORTS**

Comparing Access to In-School Predictors of Post-School Success for Students with Intellectual Disability in Conventional and College-Based Transition Services

Clare Papay, Meg Grigal, Alberto Migliore, Jie Chen, Belkis Choiseul-Praslin, and Frank Smith



 ThinkCollege

MOVING TRANSITION FORWARD

INSTITUTE FOR COMMUNITY INCLUSION, UMASS BOSTON

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Think College REPORTS present descriptive data in narrative or tabular form to provide timely information to researchers, practitioners, and policymakers. These reports provide summary data on specific elements of practice and are not intended to account for interrelationships among variables or support causal inferences. For more in-depth analyses, readers are encouraged to review other Think College publications at www.thinkcollege.net

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The Moving Transition Forward: Exploration of College-based and Conventional Transition Practices for Students with Intellectual Disability and Autism (ID/A) project examines the composition and impact of existing transition practices for students with ID/A. Through this Institute for Education Sciences (IES)-funded project, a series of secondary analysis studies were conducted with national datasets to define, explore, and compare critical aspects of two transition approaches. In this report, we look at the extent to which research-based and promising in-school predictors of post-school success are implemented in conventional transition services compared with college-based transition services for youth with ID/A. To learn more about the project, see: <https://thinkcollege.net/projects/mtf>

Read the companion report:

Matching In-School Predictors of Post-School Success to Variables in the National Longitudinal Transition Study 2012 by Clare Papay, Meg Grigal, & Alberto Migliore



Introduction

In 2004, the Individuals with Disabilities Education Act (IDEA) was expanded to require services and supports to facilitate the transition from school to post-school life for students with disabilities. After three decades of implementation, we have witnessed two subsequent reauthorizations of the law (1997, 2004); countless model demonstration projects and systems change grants; development of strong theories of practice (e.g., Kohler et al., 2016; National Collaborative on Workforce and Disability, 2019) and standards for professionals implementing those practices (e.g., Division on Career Development and Transition, 2015).

IDEA's focus on supporting transition services and supports for students with disabilities also led to an investment in understanding the transition experiences of youth with and without disabilities through the National Longitudinal Transition Study (NLTS), NLTS-2, and NLTS 2012 projects. Findings from the NLTS-2 in particular have informed much of our recent understanding of transition practices and outcomes. Over the course of nine years, the NLTS-2 project followed a nationally representative sample of students with disability who were between 13-16 years of age in 2000 as they moved from secondary school into adult roles. The primary and secondary analyses of these data have demonstrated how services and outcomes often differ for students based on their disability labels (Wagner et al., 2004) especially for those students with intellectual disability (ID) and autism spectrum disorder (ASD). Primary and secondary analysis of the NLTS-2 data has consistently shown poor post-school outcomes for students with ID and ASD (Newman et al., 2010). Students with ID had higher levels of post-school support needs than students with other disabilities and had limited participation from other agencies and support persons (e.g., vocational rehabilitation)

in their transition planning (Grigal et al., 2011; Shogren & Plotner, 2012). Similar findings are seen for students with ASD, who also had the lowest rates of employment after high school than any other group (Bouck & Joshi, 2015).

IN-SCHOOL PREDICTORS OF POST-SCHOOL SUCCESS

In 2009, Test and colleagues at the National Secondary Transition Technical Assistance Center (NSTTAC, now the National Technical Assistance Center on Transition: The Collaborative [NTACT:C]) conducted a systematic review of correlational literature on secondary transition for students with disabilities to identify in-school predictors of post-school success. Using the quality indicators for correlational research proposed by Thompson et al. (2005), Test et al. reviewed 22 correlational research studies and identified 16 in-school predictors of outcomes in the areas of employment, postsecondary education, and independent living for students with disabilities. These predictors were categorized as having a potential or moderate level of evidence based on the number and type of studies (exploratory or a priori) and consistent significant correlations between the predictor and outcome variables.

For the first time, this list of predictors included guidance on aspects of transition programming teachers and other professionals could target to support students with disabilities to achieve success after graduation. However, the list of evidence-based predictors used only the variable definitions drawn from each research study; transition professionals would need more comprehensive definitions for these predictors to be useful. To achieve this, Rowe et al. (2015) developed operational definitions and essential characteristics of each predictor through a Delphi study of experts in the field of secondary transition or career/technical education. The resulting definitions "narrow the research-to-

practice gap by giving educators information to align secondary transition programs with high quality research shown to increase the likelihood of positive post-school outcomes for youth with disabilities” (Rowe et al., 2015, p. 124).

Building on the work of Test et al. (2009) and Rowe et al. (2015), Mazzotti et al. (2016) reviewed correlational research published after 2009 and identified four new in-school predictors of post-school outcomes. The review also provided additional evidence for nine of the existing predictors. Later, Mazzotti et al. (2020) conducted another update to the review of correlational research, identifying three new predictors and providing further evidence for 14 existing predictors. At the time we began this study, there were 23 in-school predictors of post-school success, 16 of which had been operationally defined (see Table 1).

CONVENTIONAL TRANSITION SERVICES

The bulk of the correlational research reviewed to identify the predictors was based on analyses of NLTS (conducted between 1985–1990) and the NLTS-2 (conducted between 2001–2009). These studies included nationally representative samples of students with disabilities and captured the transition experiences and outcomes of students with disabilities over time using survey responses gathered from parents, students, and teachers. The NLTS and NLTS-2 were designed to gather a broad range of information on transition experiences from many students with disabilities across the United States. We view these data sources as a picture of the *conventional transition services* provided to students with disabilities as they are not designed to analyze any particular approach to transition programming (e.g., college-based transition services — see next section).

A third study in the series, the NLTS 2012, was conducted in 2012–2013 and captured data on the transition experiences of students with disabilities through interviews with students

and parents in a single school year. Phase I data from NLTS 2012 is available to researchers through a restricted use license agreement with the National Center for Education Statistics (NCES, n.d.). Phase II of the NLTS 2012 (in progress at the time of publication) will add to the available information by gathering administrative data on the secondary and post-school experiences of some of these students (National Center for Education Evaluation and Regional Assistance [NCEE], n.d.). The availability of the NLTS 2012 data provides an opportunity to closely examine the conventional transition services provided to students with disabilities using more recent data than either the NLTS or NLTS-2 datasets.

COLLEGE-BASED TRANSITION SERVICES

College-based transition services, sometimes referred to as dual enrollment or concurrent enrollment experiences, have been implemented since the late 1990s (Neubert et al., 2001). College-based transition services provide opportunities for transition-age students with intellectual disability and intellectual disability plus autism (collectively referred to here as ID/A) between the ages of 18 and 22 to receive their final years of secondary transition services in a college or university setting. As most students with ID/A continue to receive special education services until they turn 21 years old, providing transition services on a college campus allows for age-appropriate inclusion after their peers without disabilities graduate from high school (typically by age 18; Fisher & Sax, 1999). Most college-based transition programs are operated and funded by school districts, though some have been established via federal or state grants (Grigal et al., 2016; Papay & Bambara, 2011). These experiences offer students an array of campus-based activities, including access to specialized and typical college courses, engagement in job training and internships, and, in some cases, support to attain paid employment.

TABLE 1: DESCRIPTION OF IN-SCHOOL PREDICTORS OF POST-SCHOOL SUCCESS.

IN-SCHOOL PREDICTOR OF POST-SCHOOL SUCCESS	DESCRIPTION
Career awareness	Learning about opportunities, education, and skills needed in various occupational pathways to choose a career that matches one's strengths and interests
Community experiences	Activities occurring outside of the school setting, supported with in-class instruction, where students apply academic, social, and/or general work behaviors and skills
Exit exam requirements/ high school diploma status	Exit exams are standardized state tests, assessing single content area (e.g., algebra, English) or multiple skill areas, with specified levels of proficiency that students must pass to obtain a high school diploma. Diploma status is achieved by completing the requirements of the state awarding the diploma including the completion of necessary core curriculum credits.
Goal setting	Setting goals for each year (i.e., IEP goals) and adult life (i.e., post-school goals), and involving youth in development of these goals
Inclusion in general education	Access to general education curriculum and engagement in regular education classes with peers without disabilities
Interagency collaboration	A clear, purposeful, and carefully designed process that promotes cross-agency, cross-program, and cross-disciplinary collaborative efforts leading to tangible transition outcomes for youth
Paid employment/work experience	Work experience is any activity that places the student in an authentic workplace, and could include work sampling, job shadowing, internships, apprenticeships, and paid employment. Paid employment can include existing standard jobs in a company or organization or customized work assignments negotiated with the employer, but these activities always feature competitive pay (e.g., minimum wage) paid directly to the student by the employer.
Parent expectations	Parents hold high expectations for their youth, including that the youth will have a job and/or attend postsecondary education after graduation
Parent involvement	Parents/families/guardians are active and knowledgeable participants in all aspects of transition planning (e.g., decision making, providing support, attending meetings, and advocating for their child)
Program of study	An individualized set of courses, experiences, and curriculum designed to develop students' academic and functional achievement to support the attainment of students' desired post-school goals
Psychological empowerment	The belief in the relationship between one's actions and outcomes
Occupational courses	Individual courses that support career awareness, allow or enable students to explore various career pathways, and development of occupational-specific skills through instruction, as well as experiences focused on desired employment goals
Self-determination/self-advocacy	The ability to make choices, solve problems, set goals, evaluate options, take initiative to reach one's goals, and accept consequences of one's actions
Self-care/independent living skills	Skills necessary for management of one's personal self-care and daily independent living, including the personal management skills needed to interact with others, daily living skills, financial management skills, and the self-management of health care/wellness needs
Self-realization	Understanding one's strengths and support needs
Social skills	Behaviors and attitudes that facilitate communication and cooperation (e.g., social conventions, social problem solving when engaged in a social interaction, body language, speaking, listening, responding, verbal and written communication)
Student support	A network of people (e.g., family, friends, educators, adult service providers) who provide services and resources in multiple environments to prepare students to obtain annual transition and post-secondary goals aligned with their preferences, interests, and needs
Technology skills	Competence in using a computer or other technology
Transition program	Prepares students to move from secondary settings (e.g., middle school/high school) to adult-life using comprehensive transition planning and education that creates individualized opportunities, services, and supports to help students achieve their post-school goals in education/training, employment, and independent living
Travel skills	Skills that enable the student to travel to places outside of the home
Vocational education	A sequence of courses that prepares students for a specific job or career at various levels from trade or craft positions to technical, business, or professional careers
Work-study	A specified sequence of work skills instruction and experiences designed to develop students' work attitudes and general work behaviors by providing students with mutually supportive and integrated academic and vocational instruction
Youth autonomy/ decision making	Students have autonomy and make decisions about their short-term and long-range plans

Adapted from Rowe et al. (2015), Mazzotti et al. (2016), and Mazzotti et al. (2020).

The Transition and Postsecondary Programs for Students with Intellectual Disabilities (TPSID) model demonstration program, funded first in 2010 and then again in 2015 and 2020, is an initiative of the U.S. Department of Education's Office of Postsecondary Education that provides funding to institutions of higher education (IHEs) to develop or expand higher education programs for students with ID. The Higher Education Opportunities Act (2008) defined the components that should be present in TPSID programs, and, as a result, grant recipients were required to provide individual supports and services and person-centered planning for academic and social inclusion of students with ID/A in academic courses and extracurricular activities, with a focus on competitive integrated employment. Data collected from the TPSID 2010–2015 model demonstration program provide the most comprehensive national source of data on what a college-based transition experience looks like for students with ID.

PURPOSE

We sought to analyze data from the NLTS 2012 Phase I and TPSID datasets to answer the research question:

To what extent are in-school predictors of post-school success identified by Test et al. (2009), Mazzotti et al. (2016), and Mazzotti et al. (2020) reflected in the transition experiences of students with intellectual disability or intellectual disability plus autism (collectively referred to here as ID/A) in conventional and college-based transition services?

By looking for evidence of in-school predictors of post-school success, we aimed to determine the extent to which students with ID/A experience a research-based transition experience, thereby identifying areas of strength and areas of improvement in each approach to ensure students with ID/A have the best opportunity to achieve successful post-school outcomes.

Methods

DATA SOURCES

Conventional Transition Services

Data on conventional services came from the NLTS 2012 dataset. The NLTS 2012 dataset includes a national sample of about 13,000 students with and without disabilities who were in grades 7 to 12 during the 2011–12 school year. The sample included all disability categories recognized by IDEA as well as students without disabilities. The nationally representative sample was selected using a two-step process. First, researchers selected a stratified national probability sample of school districts. Of the 572 school districts selected, 76% participated in the study. Next, researchers selected a stratified sample of nearly 22,000 students with and without disabilities from the participating districts. About 11,000 students and about 13,000 parents of these students completed a computer-assisted telephone interview (48% response rate). School districts provided supplemental administrative information about the sample. More details about the research methodology of the NLTS 2012 Phase I study is available in Burghardt et al. (2017).

College-Based Transition Services

Data on college-based transition services came from a subset of a larger dataset collected from the TPSID model demonstration program. Through the TPSID model demonstration program, IHEs were awarded five-year grants to create or expand high-quality, inclusive, and model comprehensive transition and postsecondary programs for students with ID. Between 2010 and 2015, the TPSID model demonstration projects served 2,245 students, primarily between the ages of 18–25. Of this group, 445 students served were transitioning students with ID/A receiving services under IDEA (i.e., receiving college-based transition services). TPSID grantees were required to

submit annual data to the TPSID National Coordinating Center at the Institute for Community Inclusion, UMass Boston, for the purpose of evaluation. A record detailing each student's experiences in the TPSID program was completed annually. A corresponding program record was completed annually, providing information on programmatic structures and processes. Data on students who attended TPSID programs between 2010 and 2015 were analyzed for this study to align with the period of data collection for the NLTS 2012 (2012–2013).

VARIABLE IDENTIFICATION PROCESS

We identified available variables in the NLTS 2012 Phase I and TPSID datasets that corresponded with the in-school predictors of post-school success identified by Test et al. (2009), Mazzotti et al. (2016), and Mazzotti et al. (2020). We began the variable identification process by conducting a thorough review of the predictor definitions and essential characteristics (Rowe et al., 2015). We located the original research reviewed by Test et al. (2009), Mazzotti et al. (2016), and Mazzotti et al. (2020), and identified the variables in those studies that were significantly correlated with post-school outcomes. Additionally, we closely examined the variable definitions in those original research studies.

With this foundation of understanding, we next reviewed the NLTS 2012 and TPSID data dictionaries and published reports (e.g., Bloomenthal et al., 2017; Burghardt et al., 2017; Grigal et al., 2016; Lipscomb et al., 2017) to identify potential variables matching the predictor constructs. We sought variables that were either fully aligned with prior research (e.g., questions that were asked in NLTS or NLTS-2 surveys and were worded identically when asked in the NLTS 2012 parent or student surveys) or that were closely aligned with the predictor definitions and essential

characteristics (e.g., a survey question in NLTS 2012 or TPSID that was different to prior research but addressed the same construct). We conducted an iterative review process to ensure the selected the variables closely matched the predictors. This involved reviewing the data dictionary, the predictor definitions, and the original research studies to ensure our final list of variables constituted the best match. We utilized a consensus model to ensure full agreement on our final decisions.

NLTS 2012

For most of the predictors, we were able to identify a single corresponding variable from the NLTS 2012 dataset, but for one predictor (*Parent Involvement*) we identified three variables indicative of the predictor. Three predictors (*Psychological Empowerment*, *Self-Realization*, and *Youth Autonomy*) are constructs for which groups of variables in NLTS 2012 could be combined to create an index (Burghardt et al., 2017; Petcu et al., 2017; Shogren, Lee, et al., 2017). No corresponding variable in the NLTS 2012 dataset was found for eight predictors: *Career Technical Education*, *Community Experiences*, *Exit Exam Requirements/High School Diploma Status*, *Goal Setting*, *Inclusion in General Education*, *Program of Study*, *Student Support*, and *Technology Skills*. A total of 17 variables were analyzed.

TPSID

For most of the predictors, we selected a single variable from the TPSID dataset, but for four of the predictors, we decided that more than one variable was needed to provide indication of the predictor (*Paid Employment/Work Experience*, *Parent Involvement*, *Student Support*, and *Travel Skills*). No corresponding variable in the TPSID dataset was found for five predictors: *Community Experiences*, *Goal Setting*, *Parent Expectations*, *Technology Skills*, and *Transition Program*. There was insufficient data (i.e., missing data for more than two-

thirds of the sample for the corresponding variables) for five further predictors: *Exit Exam Requirements/High School Diploma Status, Psychological Empowerment, Self-Determination/Self-Advocacy, Self-Realization, and Youth Autonomy/Decision Making*. A total of 18 variables were analyzed.

SAMPLE SELECTION

NLTS 2012

In the NLTS 2012 data, primary disability is indicated by the school district, *d_y_disability*. The sample of students with ID/A consisted mainly of students who had a primary disability of ID, reported by the school district. To ensure that all students with ID were included in this group, we included students whose primary disability was ASD and whose parent responded that a professional identified the students as having ID. Therefore, this sample included students with ID/A: those who have either ID only or a dual diagnosis of ASD and ID. The sample size was $n = 1,400$ students with ID/A. We rounded all sample sizes in this report to the nearest 10, per security requirements for the NLTS 2012 dataset. Table 2 shows the demographic characteristics of these students.

Data collected from the TPSID 2010–2015 model demonstration program provide the most comprehensive national source of data on what a college-based transition experience looks like for students with ID.

TABLE 2. STUDY SAMPLE DEMOGRAPHIC CHARACTERISTICS

	NLTS 2012 %	TPSID %
Gender		
Male	59	61
Female	41	39
Race/Ethnicity		
White, Asian, or another race (non-Hispanic)	60	-
American Indian or Alaska Native	-	1
Asian	-	5
Native Hawaiian or Other Pacific Islander	-	1
White	-	67
Black (non-Hispanic)	28	21
Multi/Other	2	-
Hispanic	22	11

Note. Black includes African American; Hispanic includes Latino; Multi/Other includes more than one race selected and American Indian or Alaska Native and Native Hawaiian or Other Pacific Islander. Total *N* of the NLTS 2012 sample is 1400 students with ID/A. Total *N* of the TPSID sample is 359 students.

TPSID

From the larger TPSID 2010–2015 dataset, we extracted a sample of high school-transitioning youth receiving college-based transition services for secondary data analysis. To identify records for analysis, we first selected those in which the response to “What disabilities does this student have?” indicated the student had an intellectual disability. We then selected those records in which the response to “During this year did this student receive special education services via public school system under IDEA?” was ‘yes.’ We excluded records when the answer to this question was ‘no’ or there was no response. A total of 529 annual records were selected for analysis. Each student could have more than one annual record (meaning they attended for more than one year). We targeted the student as the unit of analysis

and aggregated annual records to create a single record for each student describing their activities across their whole college-based transition experience. The final sample consisted of 359 students (see Table 2).

DATA PREPARATION

Most variables of interest in the NLTS 2012 and TPSID datasets were dichotomous (responses of yes or no). For consistency across variables and datasets, and to mirror the methods used in previous studies (see: Papay & Bambara, 2014), we recoded all categorical variables having more than two levels of response items into dichotomous variables. See Table 3 for a list of the variables corresponding with each in-school predictor. Below, we describe steps taken to recode any non-binary variables in each dataset.

NLTS 2012

We transformed two scale ratio variables into binary categorical variables with the cutoff set as the mean for all students with IEPs reported by Lipscomb et al. (2017). Specifically, the *p_y_daily_index* variable values were split as greater than vs. equal to or below 1.46, and the *y_y_autonomy_index* variable values were split as greater than vs. equal to or below 1.63. For *psychological empowerment* and *self-realization*, we averaged seven component measures taken from the corresponding subscale of the Arc Self-Determination Scale to create an index score ranging from 0 to 1 (per methods used by Petcu et al., 2017). These scores split as greater than or equal to vs. below 0.5.

TPSID

We recoded one continuous variable (percent of student's course enrollments in inclusive courses, an indicator of the predictor "inclusion in general education") into a dichotomous variable (at least 50% vs. less than 50% of student's course enrollments were in inclusive courses).

DATA ANALYSIS

We analyzed data descriptively using SPSS v25 by calculating percentages for each variable. The following additional procedures were implemented for NLTS 2012 specifically.

NLTS 2012

We applied weights to allow for estimating the national figures based on the sample data. For the student data, we applied the weight *p_weight_allyouth*. For the parent data, we applied the weight *y_weight_allyouth*. NLTS 2012 researchers recommend these weights for analyses that do not depend on students' ages or grades. We used the SPSS Complex Samples procedure for determining accurate standard errors with weighted data (Bloomenthal et al., 2017). We rounded all figures to the closest decimal in compliance with the Institute for Education Sciences (IES) confidentiality requirements for restricted-use data.

Results

Figure 1 reflects the findings organized from high to low prevalence of in-school predictors of post-school success for students with ID. For interpretation, we divided findings into three groups according to the percentage of students with ID/A who experienced the variable: (a) highly prevalent variables (present for 70% or more of each sample); (b) moderately prevalent variables (present for between 40% and 70% of each sample); and (c) less prevalent variables (present for less than 40% of each sample).

HIGH PREVALENCE

Conventional

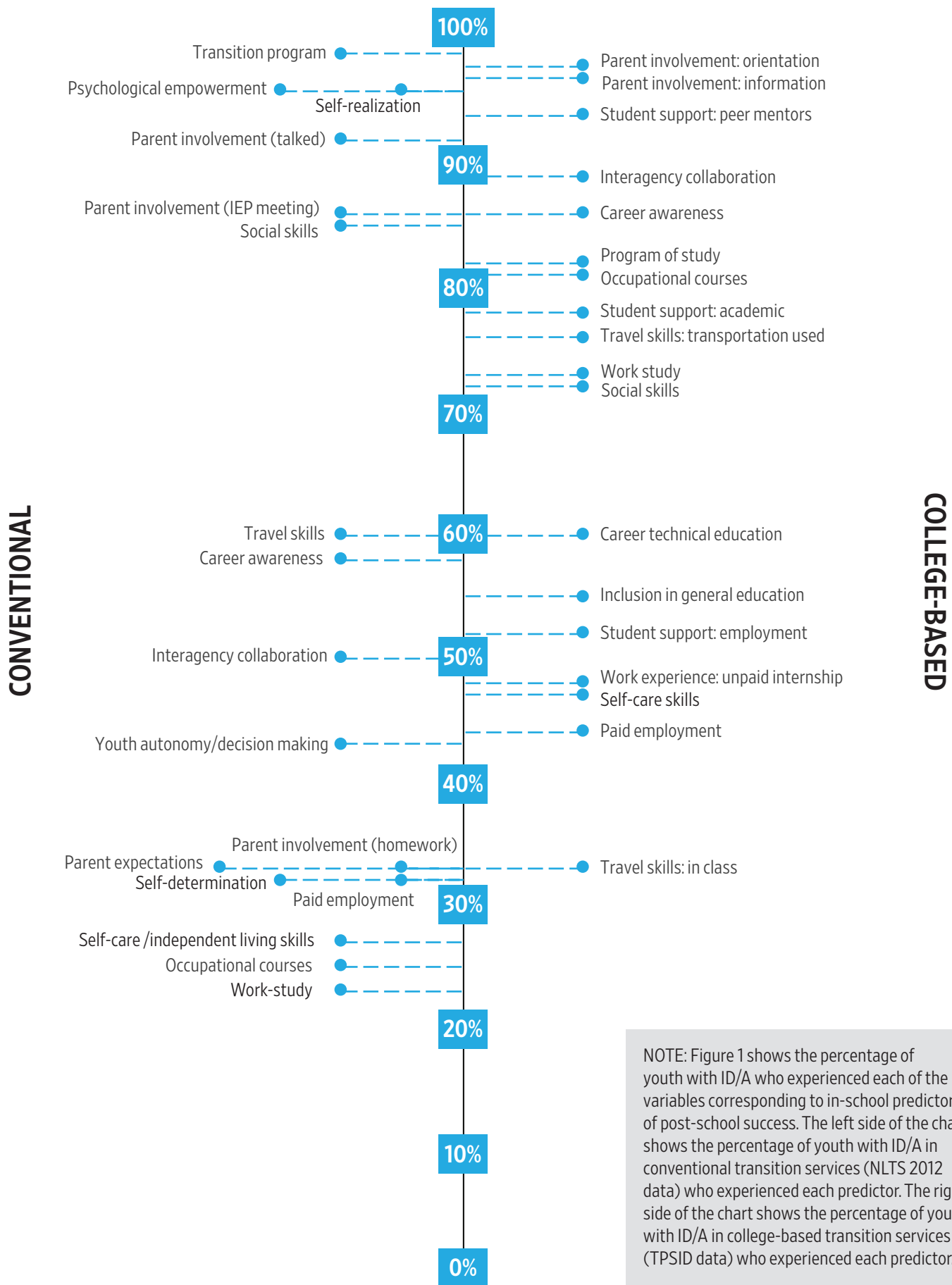
Six variables were highly prevalent (i.e., present for 70% or more students with ID) for students receiving conventional transition services. These were: (a) transition program (present for 99% of students with ID/A); (b) psychological empowerment (96%); (c) self-realization (96%); (d) parent involvement: talked about school

TABLE 3. CORRESPONDING VARIABLES WITH IN-SCHOOL PREDICTORS OF POST-SCHOOL SUCCESS

In-school predictor of post-school success	NLTS variable	TPSID variable
Career and technical education (previously vocational education)	-	Enrolled in at least two courses for which the purpose of taking the course was related to their career goals.
Career awareness	<i>K9g1</i> : Received help identifying possible career options	Participated in any unpaid/volunteer experiences
Community experiences	-	-
Exit exam requirements/ high school diploma	-	-
Goal setting	-	-
Inclusion in general education	-	At least 50% of student's course enrollments were in inclusive courses
Interagency collaboration	<i>p_y_transagency</i> : Staff from a community service agency attended the transition-planning meeting	Attended a program that had a partnership with an education agency, vocational rehabilitation (VR), community rehabilitation partner, and/or state IDD service agency (Excludes partnerships that are just project advisory committee and those that meet less frequently than every quarter)
Occupational courses	<i>F2</i> : Student took classes to prepare for a career of interest	Enrolled in at least 1 course for which the purpose of taking the course was related to their career goals.
Paid employment/ work experience	<i>y_y_anypaidjob</i> : Student had a paid work experience in the past year	Had an unpaid internship either for credit or not for credit
		Had a paid job (any type)
Parental expectations	<i>p_y_finanexp</i> : Parent expects youth will be financially self-supporting and <i>p_y_livingexp</i> : Parent expects youth will be living independently (Created a combined variable — parent expects both financial and living independence)	-
Parental involvement	<i>p_p_iepmeet</i> : (1) Parent or another household adult attended an IEP meeting in the past two years	Attended a program that provided any kind of information on transition issues to families of students attending the program
	<i>p_p_talksch</i> : (2) Parent or another household adult talked with youth about school regularly or occasionally	Attended a program that provided any kind of orientation experience to families of students attending the program
	<i>p_p_helphomework</i> : (3) Parent or another household adult helped youth with homework at least 3 times per week	-
Program of study	-	Credential was offered by program in student's first year in program
Psychological empowerment	<i>P2, P3, P4, P5, P6, P7, P8</i> : Student's rating of self on questions based on the Arc's Self-Determination Scale subsection on Psychological Empowerment	-

In-school predictor of post-school success	NLTS variable	TPSID variable
Self-care/ independent living skills	<i>p_y_daily_index</i> : Activities of daily living index	Enrolled in at least 1 specialized course during their college-based transition services experience that taught independent living
Self-determination/ self-advocacy	<i>p_y_goals</i> : Student played at least an equal part in developing plan goals	-
Self-realization	<i>P9a, b, c, d, e, f, g</i> : Student's rating of self on questions based on the Arc's Self-Determination Scale subsection on Self-Realization	-
Social skills	<i>p_y_communicate</i> : Student communicates by any means with little or no trouble	Enrolled in at least 1 specialized course during their college-based transition services experience that taught social skills
Student support	-	(Academic) Enrolled in a class and received academic supports from an individual
	-	(Employment) Had a paid job and received employment supports from an individual
	-	(Peer mentors) Attended a program that provided peer mentor supports
Technology skills	-	-
Transition program	<i>E3</i> : Transition plan meeting by youth's school occurred	-
Travel skills	<i>D26c</i> : Student can get to places outside home very well or pretty well	Enrolled in at least 1 specialized course during their college-based transition services experience that taught travel skills
		Used transportation options (walked, drove self, rode bike, took public transportation, took paratransit, took taxi, or took campus transportation) to get to class and campus activities in at least one year they attended the program
Work-study	<i>y_y_schjob</i> : Student had a paid or unpaid school-sponsored work activity in the past year	Enrolled in at least one course related to their career goals AND either had a paid job or did unpaid career development
Youth autonomy/ decision making	<i>y_y_autonomy_index</i> : Personal autonomy index	-

FIGURE 1. PREVALENCE OF IN-SCHOOL PREDICTORS OF POST-SCHOOL SUCCESS IN CONVENTIONAL VS. COLLEGE-BASED TRANSITION SERVICES FOR STUDENTS WITH INTELLECTUAL DISABILITY



(92%); (e) parent involvement: attended IEP meeting (86%); and (f) social skills (85%).

College-Based

Eleven variables were highly prevalent for students receiving college-based transition services. These were: (a) parent involvement: orientation (98%); (b) parent involvement: information (97%); (c) student support: peer mentors (94%); (d) interagency collaboration (89%); (e) career awareness (86%); (f) program of study (82%); (g) occupational courses (81%); (h) student support: academic (78%); (i) travel skills: transportation used (76%); (j) work study (73%); and (k) social skills (72%).

MODERATE PREVALENCE

Conventional

Four variables were moderately prevalent (i.e., present for between 40% and 70% of students with ID/A) for students receiving conventional transition services: (a) travel skills (60%); (b) career awareness (58%); (c) interagency collaboration (50%); and (d) youth autonomy/decision making (43%).

College-Based

Six variables were moderately prevalent for students receiving college-based transition services: (a) career/technical education (60%); (b) inclusion in general education (56%); (c) student support: employment (52%); (d) paid employment/work experience: unpaid internship (48%); (e) self-care skills (47%); and (f) paid employment/work experience: paid job (44%).

LOW PREVALENCE

Conventional

Seven variables were less prevalent (i.e., present for fewer than 40% of students with ID/A) for students receiving conventional transition services: (a) parent involvement: helped with homework (33%); (b) parent expectations (33%); (c) paid employment/work experience

(32%); (d) self-determination/self-advocacy (32%); (e) self-care/independent living skills (27%); (f) occupational courses (25%); and (g) work study (23%).

College-Based

One variable was less prevalent for students receiving college-based transition services: Travel skills (specialized classes; 33%).

Discussion

Variations in quality of transition services provided to students with ID/A have led to long-term inadequacy and inequality in these students' education and employment outcomes (Bouck, 2012; Bouck & Joshi, 2015; Grigal et al., 2011). Students with ID/A have lower rates of employment and earn lower wages, and their enrollment in postsecondary education lags their peers with other disabilities (Carter et al., 2012; Grigal et al., 2014). Most students with ID/A exit high school and enter a lifetime of sheltered employment (earning subminimum wage) or day habilitation (Siperstein et al., 2013).

Test et al. (2009), Mazzotti et al. (2016), and Mazzotti et al. (2020) have led efforts to understand *what works* to achieve better post-school outcomes in employment, education, and independent living by identifying in-school predictors of post-school success in studies using the NLTS-2 dataset and other sources. Data from these studies were critical to our understanding of students' transition experiences and how they shape post-school outcomes. However, these data also reflect practices and student experiences from up to 20 years ago. The more recently collected data presented in the NLTS 2012 Phase I dataset offered an opportunity to examine the extent to which these in-school predictors of post-school success are evident in our most current national conventional transition data for students with

ID/A. Furthermore, the availability of the subset of TPSID data reflecting the practices used in college-based transition programs for the first time offers an opportunity to examine which of these in-school predictors of post-school success are evident in college-based transition programs as well as a chance to contrast these two transition approaches. In the following sections, we compare the prevalence of variables matching the in-school predictors of post-school success identified in each dataset.

COMPARING CONVENTIONAL AND COLLEGE-BASED TRANSITION SERVICES

We identified 17 variables in the NLTS 2012 dataset and 18 variables in the TPSID dataset for analysis. For conventional transition services, we were able to identify variables in NLTS 2012 matching 15 of the research-based predictors of post-school outcomes. For college-based transition services, we were able to identify variables matching with 13 of the research-based predictors. Because a few predictors had multiple matching variables (three variables matching *Parent Involvement* in NLTS 2012; three variables matching *Student Support*, two variables matching *Paid Employment/Work Experience*, two variables matching *Travel Skills*, and two variables matching *Parent Involvement* in the TPSID dataset), the total number of variables for analysis was close to the same in each dataset. Visual analysis of Figure 1 shows the results fell into three levels of prevalence based on the percentage of students with ID/A who experienced each predictor: high, moderate, and low. College-based transition services had more variables in the high and moderate prevalence ranges for students with ID/A, and conventional transition services had a higher number of variables in the low prevalence range.

In the next sections, we reflect on the comparison between conventional and college-based transition services via the variables

in each prevalence group. It is important to note that high level of prevalence does not indicate these predictors were more or less important or impactful than other predictors. The predictors offered by Test et al. (2009), Mazzotti et al. (2016), and Mazzotti et al. (2020) each have varying levels of evidence to support their correlation with positive outcomes, but they have yet to be subjected to experimental analysis to determine which predictors are more likely to lead to positive post-school outcomes.

High Prevalence

Variables present for more than 70% of students with ID/A in each dataset were deemed as being highly prevalent. Eleven variables were highly prevalent in college-based transition services compared with six variables highly prevalent in conventional transition services. It should be noted we did not identify specific variables in the college-based transition services data reflecting the research-based predictors *Community Experiences* and *Transition Program*. However, all the students with ID/A included in the data were participating in a transition program on a college campus, which is, in fact, a community experience. Therefore, 100% of the students receiving college-based transition services were engaging in community experiences and in a transition program. But since these elements of their experiences were not specifically aligned with variables in the dataset, we did not include them in our calculations.

Two of the highly prevalent variables identified in the conventional transition services reflected compliance with federal statute: the existence of a transition program and parent involvement.

College-based transition services had more variables in the high and moderate prevalence ranges for students with ID/A, and conventional transition services had a higher number of variables in the low prevalence range.

We used the variable “a transition plan meeting by youth’s school occurred” to support the existence of a transition program. Given that transition programs are a federal requirement, it is not surprising that almost all students with ID/A receiving conventional transition services experienced this variable. Similarly, the law requires parent involvement — for example, in a child’s IEP meeting or talking about school. Therefore, these two highly prevalent variables could be considered baseline in-school practices. Parent involvement variables were also highly prevalent in college-based transition experiences. In this case, the variables aligned with parent involvement in orientation activities and the delivery of information to parents during the program. These two activities are not mandated as part of IDEA but reflect the opportunities college-based transition services programs offered to help parents understand the goals, activities, and outcomes of their program during orientation. This information sharing was maintained throughout the course of student’s transition experience.

The variable identified as an indicator of social skills (“student communicates by any means with little or no trouble”) was highly prevalent (85%) for students with ID/A receiving conventional transition services. This is encouraging because student social skills have been found to increase students’ chance of employment after school (Carter et al., 2012). It is further encouraging that when using the criteria we set for *psychological empowerment and self-realization* (student’s average of seven component measures for each of these was at least 0.5), almost all students with ID/A were determined to have experienced these important predictors of post-school outcomes.

Previous studies have established the aim of college-based transition services is to address student’s academic, employment, and social goals in a college setting (Schillaci et al., 2021).

The highly prevalent variables observed in the college-based transition services dataset reflect the emphasis placed on these areas in practice. Career awareness, occupational courses, student academic support, work-study, travel skills (transportation used), and social skills all were highly prevalent. Students with ID/A in college-based transition services programs enrolled in college coursework, creating an academic experience that would likely not be available if the transition services were received in a high school. The presence of three highly prevalent variables associated with employment (work study, occupational courses, career awareness) demonstrates that many students with ID/A in college-based transition services programs were engaged in employment preparation activities. Career awareness in TPSID supported college-based transition programs included a variety of activities, such as providing access to job coaches/developers; offering internships, service-learning opportunities, and paid work experiences; and connecting with service providers to sustain employment (Grigal et al., 2017).

The remaining highly prevalent predictors included student support provided by peer mentors, interagency collaboration, travel skills, and program of study. Peer mentors are widely used in higher education programs to offer academic and social support (Grigal et al., 2021).

Mentors can be paid or unpaid and receive training and oversight from program staff. These interactions are reciprocally beneficial, offering the students with ID/A access to informal academic and social coaching to better navigate the college expectations. Peers have also reported positive outcomes, including building more positive self-identity, improved civic engagement, and advocacy (Athamanah, 2020).

Interagency collaboration, another highly prevalent variable in college-based transition, was defined as “attended a program that

Students' use of various transportation modalities demonstrated that college-based transition programs offered students substantial opportunity to access and use an array of travel options.

had a partnership with an education agency, vocational rehabilitation (VR), community rehabilitation partner, and/or state IDD service agency.” The unique nature of college-based transition programs requires the hosting college or university to, at a minimum, collaborate with the local education agency sending the students being served. However, these programs also collaborated with other state agency personnel. Published reports on the full universe of the TPSID program data from the same period reflect high levels of collaboration with VR and IDD state agencies as well as others external partners, including community rehabilitation providers and employers (Grigal et al., 2017).

Students' use of various transportation modalities demonstrated that college-based transition programs offered students substantial opportunity to access and use an array of travel options. Students could drive themselves, walk, or ride a bike; take public transportation/paratransit or taxis; or use campus transportation. Seventy-six percent of students attending college-based transition programs used one or more of these options to get to class and campus activities in at least one year they attended the program. Use of public transportation was most prevalent (64%); followed by driving self/walking or riding bike (25%); and using campus transportation (14%). Previous studies have mentioned college-based transition programs offer training to support students' use of public transportation (Grigal & Dwyre, 2010; Rogan et al., 2014). Some students use it to get to and from the campus,

and, in some cases, their jobs. These data show college-based transition programs offer students a chance to build their travel skills using an array of transportation options, which will likely be useful as they exit their program and enter the workforce.

Finally, the presence of a program of study — defined for the college-based transition programs as “credential was offered by program in student's first year in program” — shows most of the students with ID/A enrolled in college-based transition programs had a plan guiding their academic, career, and social experiences that would lead them to an identified credential. Students attending TPSID programs have a high rate of credential attainment; in 2015 (the final year of the first cohort), over 80% of students attained one or more credentials (Grigal et al., 2017).

Moderate Prevalence

Variables present for between 40% and 70% of students with ID/A in each dataset were deemed as being moderately prevalent. Six variables were moderately prevalent in college-based transition services, compared with four variables found to be moderately prevalent in conventional transition services. Four of the six moderately predictors evident in college-based transition programs related to employment, including: career and technical education; student support for employment; paid employment/work experience: unpaid internship; and paid employment/work experience: paid job. Roughly half of the students enrolled in college-based transition programs were being provided with unpaid and paid internships, and slightly fewer had paid jobs while enrolled. Previous studies of TPSID data have found students with paid jobs are 15 times more likely to have paid job at exit (Grigal et al., 2019); therefore, these preparation activities can have a strong impact on student employment outcomes.

One employment related variable was moderately prevalent in conventional transition services: career awareness. Defined as “job seeking instruction,” career awareness was evident for more than half (58%) of students with ID/A receiving conventional transition services. This is encouraging, given its indicator status as predictive of post-school employment. However, it is somewhat concerning this essential practice was not more highly prevalent and no other employment-related variables were found to be moderately prevalent.

The predictor “inclusion in general education” was matched in the college-based transition dataset with this variable: at least 50% of student’s course enrollments were in inclusive courses. Moderate prevalence of access to traditional college courses in college-based transition programs aligns with previous findings on the fuller realm of higher education programs enrolling students with ID/A. Many higher education programs enrolling students with ID/A offer a mix of both inclusive and specialized courses (Grigal et al., 2022). The rate of inclusion for students in college-based transition programs (56%) appears to be higher than it was for the general population of college students with ID/A in the TPSID dataset, which was under 50% during this time. Access to inclusive college courses is a central tenet of the TPSID program as it allows students to potentially earn credits, access meaningful credentials, and participate in learning experiences with other college students. However, other college-based programs may not have the same level of access to inclusive courses.

There were no variables in the NLTS 2012 reflecting the level of inclusion in the conventional transition services dataset. Since there are no means for comparison, we cannot purport there is more inclusion in general education in college-based transition

services. What we can confirm is that inclusion in general education is better documented in college-based transition services data than in conventional transition services data.

The other moderately prevalent variables in conventional transition services were travel skills; interagency collaboration; and youth autonomy/decision making. Travel skills — defined as “youth being able to get to places outside home very well or pretty well” — were prevalent for 60% of students with ID/A. These findings are slightly different from the findings of Roux et al. (2020), who conducted a recent study on high school work experiences of students with ASD and ID using the NLTS 2012 dataset. They found half of students with ID were able to navigate to places outside the home “very well,” compared to one-third of students with ASD. This discrepancy may be due to the sample selection process. In the Roux et al. study, students with ASD and co-occurring ID were included in the ASD group only, whereas in our study, students with ID with co-occurring ASD were classified as having ID (referred to as ID/A here). The level of overlap between these groups of students in the NLTS 2012 dataset may lead to future difficulties in interpreting study findings.

Interagency collaboration, a highly lauded practice in transition services, is often operationalized by the involvement of adult agencies in education planning or service provision. Interagency collaboration has been found to be a predictor of post-school education and employment (e.g., Mazzotti et al., 2020). However, previous studies examining interagency collaboration for students with ID/A have documented relatively low levels of involvement from outside agency staff in IEP planning meetings. For example, in an analysis of transition planning for students with ID, ASD, or other disabilities using NLTS-2 data, Shogren and Plotner (2012) found only 23% of students

Schillaci et al. (2021) found one year of participation in a college-based transition program led to significant increases in students' autonomy, self-initiation, psychological empowerment, and self-realization.

with ID had an actively-involved vocational rehabilitation counselor, and only 15–16% of students with ID or ASD had active participation from another outside agency. Our findings indicate interagency collaboration is more prevalent in the NLTS 2012 data set: 50% of students with ID/A had staff from a community service agency attend their transition-planning meeting. This demonstrates an improvement in collaboration with community service agencies in conventional transition services; however, this practice is still substantially less prevalent than the 89% of students enrolled in college-based transition programs receiving interagency collaboration.

Youth autonomy/decision-making was also moderately prevalent in conventional transition services, but the way this variable was defined for analysis should be noted. To create a binary variable in line with other variables in the analysis, we defined youth autonomy/decision making to be “present” if the youth scored at or above the average that was reported for all youth on this scale by Lipscomb et al. (2017) — in other words, a score of at least 1.63 on the *y_y_autonomy_index* variable. Forty-three percent of students with ID/A scored at least 1.63 on this scale, meaning 57% scored below the average that was reported for all students with disabilities by Lipscomb et al. (2017). This suggests a substantial portion of students with ID/A have lower levels of autonomy. In a study using NLTS-2 data, Shogren, Garnier Villarreal, and colleagues (2017) found autonomy, combined with psychological empowerment and self-realization, played a significant role in mediating the relationship between school-

based student, family, and school factors and post-school outcomes. Other studies have also documented the direct and indirect impact of self-determination on school factors and student outcomes (e.g., Shogren, Garnier Villarreal, et al., 2017).

Variables on youth autonomy and decision making were not present in the college-based transition services data, so no comparison is possible between the two approaches. Two previous studies shed some light on these and other constructs related to self-determination and how they may be impacted by a college-based experience. A study conducted by Shogren et al. (2018) used data on first-year students enrolled TPSID cohort 1 programs and found the number of social activities students participated in predicted their scores on autonomy, self-realization, and overall self-determination. They suggested participating in social activities potentially supports student self-awareness and self-realization but also helps enhance choice and access to preferred activities (autonomy). Another study, by Schillaci et al. (2021), explored the impact of college-based transition services on self-determination for youth with intellectual and developmental disabilities. They found one year of participation in a college-based transition program led to significant increases in students' autonomy, self-initiation, psychological empowerment, and self-realization. Self-determination and youth autonomy remain a critical area of concern for transition-age students with ID regardless of the transition approach.

Low Prevalence

Variables present for fewer than 40% of students with ID/A in each dataset were deemed as having low prevalence. Low prevalence was seen with variables in conventional transition services corresponding to seven predictors (*Parent Involvement:*

Helped with Homework; Parent Expectations; Paid Employment/Work Experience; Self-Determination/Self-Advocacy; Self-Care/Independent Living Skills; Occupational Courses; and Work-Study). It is particularly concerning that three of the seven variables reflecting low prevalence for students with ID/A were in the employment domain (*Paid Employment/Work Experience, Occupational Courses, and Work Study*). Previous research has repeatedly established that early employment in high school predicts employment post high school (Gold et al., 2013; Test et al., 2009; Wehman et al., 2015). The low prevalence of paid employment (32%) demonstrates conventional transition practices are not sufficiently or consistently providing employment preparation activities for most students with ID/A. These study findings offer discouraging news about schools' transition programs using practices which lead to more successful post-school employment for students with ID/A.

The variable we analyzed as an indicator of *Parent Expectations* pulled from two items: "parent expects youth will be financially self-supporting" and "parent expects youth will be living independently." Given the low levels of paid employment found for young adults with ID/A, these data may merely reflect pragmatism of the parents surveyed. Data from the National Core Indicators Project suggest that in 2014, only 4% of 18–21-year-old students supported by state intellectual or developmental disability (IDD) agencies were employed, and only 9% of those between 22–30 years old were employed (Butterworth & Migliore, 2015). However, the low prevalence of high parental expectations in conjunction with the low prevalence of work-related predictor variables may indicate a system predisposed toward offering transition experiences not reflective of high expectations for positive employment outcomes.

The one area of low prevalence in the college-

based transition dataset was the second variable related to travel skills: "students were enrolled in at least one specialized course during their college-based transition services experience that taught travel skills." A key phrase in this variable is *specialized course*, meaning a course only attended by TPSID students (i.e., not inclusive). A foundational tenet of the TPSID programs was to support access to inclusive (not specialized) courses; therefore, this finding may be more indicative of where and when travel instruction was provided, not that it wasn't available. The other variable reflective of travel skills discussed earlier (transportation used) demonstrates that the majority of students used various forms of transport to get to campus and classes.

One final point worthy of reflection is about how the nature of each dataset impacted the variables applicable to our analyses. The NLTS 2012 Phase I involved surveys of parents and students. As such, it provided information on parent expectations and involvement as well as students' perspectives on their self-advocacy, psychological empowerment, and self-realization. What it lacked, however, was information on transition programming from the perspective of those planning and delivering transition services. No information was provided on aspects such as the student's program of study, if or how they set goals, or if they learned technology skills or focused on career/technical education. Conversely, the subset of the TPSID data were reported entirely by transition program staff. As such, this dataset provides a rich source of information on transition programming, including the academic and employment experiences of students in college-based transition services and whether programs offered supports for parents and students. But these data lacked information drawn directly from parents and students on their expectations, goals, and self-determination. Interestingly, of the research-based predictors

that we were not able to match up with variables, four predictors had no match in either dataset (*Community Experiences*, *Exit Exam Requirements/High School Diploma Status*, *Goal Setting*, and *Technology Skills*), suggesting a lack of attention to these constructs in large scale transition studies.

Limitations

Several limitations should be noted about this study. First, the way students were categorized as having ID vs. ASD may impact interpretation of our findings and how they compare with findings from other studies. As mentioned earlier, other researchers have categorized overlaps between these student groups in different ways (see Roux et al., 2020). In our study, the category of students with ID included students with ASD who had also been identified as having ID. Students with ASD who did not have a dual diagnosis of ID were categorized as having ASD. In contrast, Roux et al. (2020) included students in the ID group if they received special education services under the ID category, or if they had a parent-reported ID diagnosis and were receiving special education services under a category other than ID. Roux et al. (2020) excluded students from the ID group if they had parent-reported co-occurring ASD, or if they received special education services under the ASD category. Students with ASD and co-occurring ID were included in the ASD group only. It is important to note this difference in grouping when making comparisons between studies.

Another limitation of our study was the inability to identify variables in the NLTS 2012 and TPSID datasets for each in-school predictor of post-school success. Without these data, it is impossible to determine the prevalence of these predictors in our most current national datasets on transition-age students in the U.S. Our approach of seeking at least one

variable corresponding to each of the in-school predictors of post-school success provides an initial look at the extent to which students with ID/A are accessing research-based transition experiences, but this is an incomplete picture. Transition experiences comprise more than what can be captured in a single variable connected with each important predictor, and future studies may be able to extend this work by examining additional data.

In addition to the gaps presented by unmatched predictors, some of the variables selected were not perfect matches but were close matches given the available data. A good example of this was *Social Skills*, which Rowe et al. (2015) defined as “behaviors and attitudes that facilitate communication and cooperation” (e.g., social conventions, social problem solving when engaged in a social interaction, body language, speaking, listening, responding, and verbal and written communication). This predictor was identified based on six studies showing a positive relationship between higher social skills and successful post-school outcomes (e.g., Halpern et al., 1995). Although the NLTS and NLTS-2 included several questions for parents about their student’s social skills, NLTS 2012 did not include these questions. The closest approximation we could locate was the variable *p_y_communicate*, which provides an indication of how well the student communicates by any means by combining variables related to speaking (for students who do not have issues with hearing or speaking) and communication by other means (for students who have issues with hearing or speaking).

Although not a perfect match, the *p_y_communicate* variable was similar to the NLTS-2 variable used by Carter et al. (2012) as an indicator of social skills. The Carter et al. (2012) study identified social skills as a predictor of post-school outcomes; thus, we felt using it in the present study was supported by the

literature. In the TPSID dataset, we were able to identify a variable providing information on whether students took a specialized (i.e., separate) class that covered the subject “social skills training.” This provides an indication of whether students received some instruction in social skills (as opposed to understanding their level of social skills). However, it likely underestimates the extent to which students with ID/A in college-based transition services are taught social skills, because it captures only instruction provided in separate classes and not instruction that might take place in natural and/or community-based settings.

Implications

Despite these limitations, the findings from this study contribute to understanding the prevalence of practices which reflect in-school predictors of post-school success for students with ID/A in conventional and college-based transition services. NLTS 2012 is the best available dataset for exploring the implementation of research-based transition services across the U.S. The sample we analyzed from this dataset was large and represented the national population of transition-age students with and without disabilities. The TPSID dataset, while not a representative sample, offers the first cross-program and cross-state dataset reflecting college-based transition programming for students with ID/A. Although relatively small, this dataset provides valuable information about how college settings offer unique opportunities for practitioners to implement research-based transition practices. The findings of this study also identify potential advantages to offering college-based transition services, as well as guidance about priority areas where transition services could be improved in both conventional and college-based approaches.

Early paid work experiences while enrolled in high school have repeatedly been found to be a strong predictor of post-school competitive employment (Test et al., 2009).

IMPLICATIONS FOR PRACTICE

Within conventional transition services in particular, the low prevalence of predictor variables related to employment (e.g., paid employment/work experiences, occupational courses, work study) has direct implications for transition educators currently working with students with ID and ASD in secondary education. To prepare for employment, individuals with ID and ASD require work experiences during high school. Experiences may include both in-school and out-of-school work and paid and unpaid experiences. Students also benefit from inclusion in occupational courses and work study classes. Flexer et al. (2011) found students with ID who completed three or more semesters of career and technical education were more likely to be working one year after exit. Early paid work experiences while enrolled in high school have repeatedly been found to be a strong predictor of post-school competitive employment (Test et al., 2009). The transition field must prioritize supporting students with ID and ASD to seek and maintain paid employment while enrolled in high school and to engage in existing work study and occupational courses available in high schools. Given that employment related preparation activities were more prevalent in the college-based transition program data, efforts could be made by existing high school transition programs to explore how employment services are being planned for, staffed, and implemented in college-based programs to better inform their own practices.

Further, given the higher prevalence of research-based practices in college-based transition programs overall, local education agencies could look to cultivate more of these opportunities for their students with ID/A. A strong example of an effective initiative has been implemented in Massachusetts since 2007. The state has funded the development of 15 *partnerships between local school districts and two- and four-year public colleges and universities in Massachusetts* to support transitioning youth (Massachusetts Department of Higher Education, 2022). *Participating students are included in the campus community as they develop their capacities in career-planning and employment, self-advocacy, and other self-determined abilities that support their post-secondary interests and activities. Expansion of college-based transition services would provide students with ID/A more opportunity for academic and social inclusion with same-age peers as well as access to employment preparation and work opportunities less available in conventional transition services.*

In conventional transition services, it was affirming to see the high prevalence of psychological empowerment and self-realization predictors for youth with ID/A. However, it is difficult to reconcile these findings with the relatively low prevalence of self-determination and self-advocacy found for students with ID and ASD. To enhance students' growth in self-determination, transition educators could explore the use of effective models such as the Self-Determination Learning Model of Instruction (Shogren, Wehmeyer et al., 2017), which can be implemented in a variety of classroom contexts across a range of core content areas to enhance student capacity to set goals related to learning core content, developing actions plans, and evaluating their progress toward those goals.

While not present as a variable in the TPSID

dataset, others have documented the impact college-based transition services have on the development of self-determination for students with ID/A (Cook et al., 2021; Schillaci et al., 2021). These transition experiences also offer students the opportunity to set goals, choose classes, navigate campuses and places of employment, and participate in social activities (Cook et al., 2021). Given how strongly these activities align with in-school predictors of post-school success, future development of college-based transition practices is worthy of consideration.

IMPLICATIONS FOR RESEARCH

Our findings reflect the prevalence of in-school predictors of post-school success in conventional transition services is not optimal. One question that future research should try to address in the future is: **Why?** Why are so few students with ID/A receiving employment preparation as part of their transition experiences? Why are so few students with ID/A perceived as not possessing self-determination skills and self-advocacy skills? Why are parent expectations for the child's future financial success and independent living so low? Our knowledge about what works (i.e., the research-based predictors) is becoming clearer. Researchers continue to define, investigate, and articulate the practices needed to successfully support students with ID/A to achieve positive post-school outcomes (Test et al., 2009; Rowe et al., 2015; Mazzotti et al., 2016; Mazzotti et al., 2020). What remains unknown is why these effective practices are not used consistently in the field.

Are practitioners limited by administrative structures? Are transition personnel and secondary general and special education teachers not trained in the use these practices? Or is the special education transition system not prioritized by the administrators in local education agencies charged with implementing

these services? Given the broad population of students represented by the NLTS 2012 dataset, the low prevalence of research-based practices for students with ID/A is highly concerning. Better answers are needed to inform future policy development and ensure better experiences while enrolled in high school as well as better outcomes after exiting high school.

We were unable to locate matching variables for eight in-school predictors of post-school success in the NLTS 2012 dataset. Many of these variables were present in previous NLTS datasets, and significant gaps may emerge in future research without updated data on these variables. Given the extent of missing data and the absence of outcome data, it may be impossible to pursue replication of some of the correlational studies done on NLTS-2 data with NLTS 2012 data. As future datasets emerge from Phase II NLTS collections, researchers will need to highlight when these gaps have been addressed and clarify how the more current data compare with the previous datasets.

The college-based transition subset of the TPSID data offers the only available national data describing practices used in college-based transition programs for students with ID/A, but these data only reflect practices used in federally funded programs. Future research needs to identify a broader and more representative sample. Data on program availability and transition services implementation would allow for a deeper understanding of this approach to transition service delivery. Additionally, the collection of follow-up data providing information about students' employment and independent living status after exiting special education would provide critical information that could be compared with outcome data from other datasets. If these data were gathered consistently and updated annually, research

could be conducted on other research-based predictors, allowing the field to document how certain aspects of the program impact, and perhaps predict, desired outcomes.

Finally, as future longitudinal studies are designed, we recommend researchers consider aligning their data collection efforts with these predictors and, when possible, the operational definitions provided by Rowe et al. (2015). This could lead to greater consistency in terminology and clarity when comparing study findings. To further extend our findings, future research on the predictors could focus not only on their prevalence, but also their impact through measurement of post-school outcomes.

Conclusion

The emergence of new datasets on transition-age students offers renewed opportunities to assess if the practices deemed effective by previous research are evident in these data. By mapping the in-school predictors of post-school success onto variables in the NLTS 2012 Phase I and a subset of the TPSID dataset, we were able to identify the extent to which students with ID/A experienced these important predictors of post-school success in conventional and college-based transition services. We found in-school predictors of post-school success were more highly prevalent in college-based transition services than in conventional transition services. Given the strong prevalence of research-based practices evident in college-based transition programs, this method of transition service delivery is worthy of continued development and study. However, our findings are not able to verify if college-based transition services offer a structure more representative of post-school success. This question remains to be answered through future research. To do this, we need increased availability of college-based transition services as a transition option for students with

ID/A across the country. We also need stronger, more comprehensive datasets to enable us to understand the transition experiences of students with ID/A regardless of how they receive their transition services. With better and more granular data, future researchers can explore which research-based predictors are *most* important for students with ID/A and why, thereby allowing practitioners to understand how to deploy the most effective transition services to ensure successful post-school outcomes for all students.

We found in-school predictors of post-school success were more highly prevalent in college-based transition services than in conventional transition services.

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